INTRODUCTION TO GENERAL CHEMISTRY 1



WHAT ARE YOU MADE OF?

YOUR NAME

YOUR STRENGTH(S)

YOUR WEAKNESS(ES)

YOUR MOTIVATION(S)

YOUR SHORT TERM GOAL(S)

YOUR LONG TERM GOAL(S)

Learning Objectives:





Compare and contrast various states of matter (solid, liquid, gas, plasma) in terms of shape, volume, compressibility, and particle movement.

Show appreciation for the role of matter and its properties in daily life and various fields (e.g., medicine, engineering, environmental science).

Perform comparative tests (e.g., heating different substances to observe changes in state or color) and identify whether changes are physical or chemical.





WHAT IS GENERAL CHEMISTRY?





- change in matter
- intro to all areas of chemistry
- laboratory environment
- experiments





WHY IS IT IMPORTANT?





- deeper understanding
- know the composition of drugs or medicine
- chemical properties
- sterilizing and sanitation
- diagnosis of disease
- discover new drugs
- how the drug acts







GENERAL CHEMISTRY 1



"khēmeia" = transmutation (change of form) of Earth.







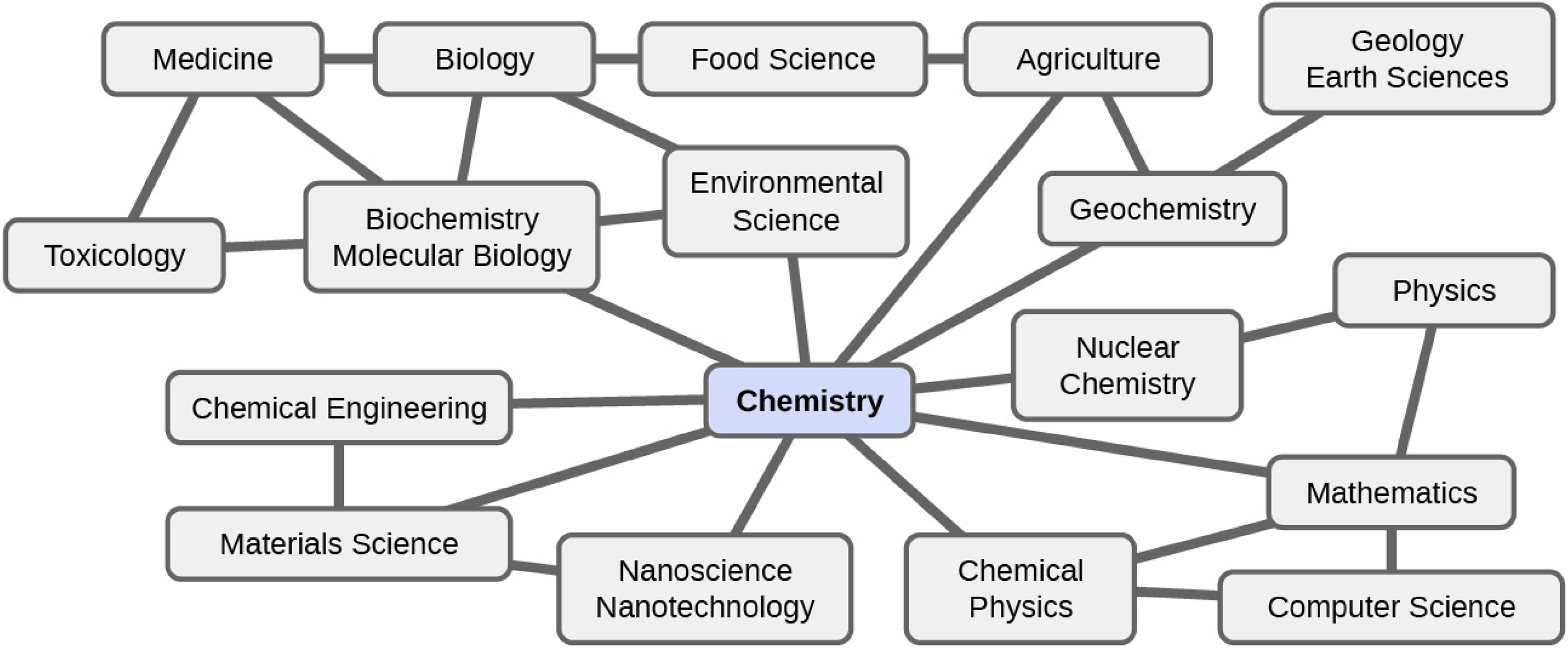
LEARNING IN ACTION



Think of at least 10 careers/jobs/fields where General Chemistry is useful and explain in what way is GenChem utilized in these fields.






Medicine
Biology
Food Science
Agriculture
Geology
Earth Sciences
Toxicology
Biochemistry
Molecular Biology
Environmental
Science
Geochemistry
Physics
Nuclear
Chemistry
Chemical Engineering
Chemistry
Mathematics
Materials Science
Nanoscience
Nanotechnology
Chemical
Physics
Computer Science

1. **Analytical chemistry** uses qualitative and quantitative observation to identify and measure the physical and chemical properties of substances. In a sense, all chemistry is analytical.
2. **Physical chemistry** combines chemistry with physics. Physical chemists study how matter and energy interact. Thermodynamics and quantum mechanics are two of the important branches of physical chemistry.
3. **Organic chemistry** specifically studies compounds that contain the element carbon. Carbon has many unique properties that allow it to form complex chemical bonds and very large molecules. Organic chemistry is known as the "Chemistry of Life" because all of the molecules that make up living tissue have carbon as part of their makeup.
4. **Inorganic chemistry** studies materials such as metals and gases that do not have carbon as part of their makeup.
5. **Biochemistry** is the study of chemical processes that occur within living organisms.

- To know the composition of drugs or medicines. eg. Drug contents
- To understand the chemical properties of the drug. Eg. Alkaline or acidic, water soluble or fat soluble.
- Predicting the drug interaction. Eg. Reaction of an antibiotic with an antacid given for ulcer.
- For sterilizing and sanitation. Eg. Alcohol, phenol, acids used for cleaning.
- In diagnosis of disease. Eg. Use for glucose strips for checking blood sugar.
- To regulate drug distribution. Eg. High dose of as drug which makes blood acidic can be stabilized by an recommended alkaline drug.
- Decrease the toxic effect.
- To discover new drugs with better effectiveness.
- To study the mechanism of disease.
- To know how the drug acts.



INTRODUCTION TO MATTER



- anything that takes up space and has mass
- made of molecules
- composed of atoms
- molecules are either moving or in stasis

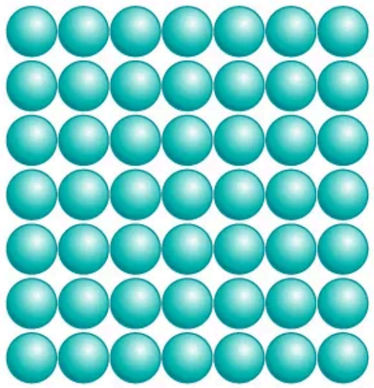




STATES OF MATTER

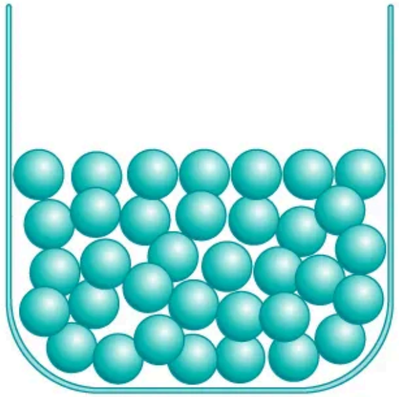
Physical states

increasing energy



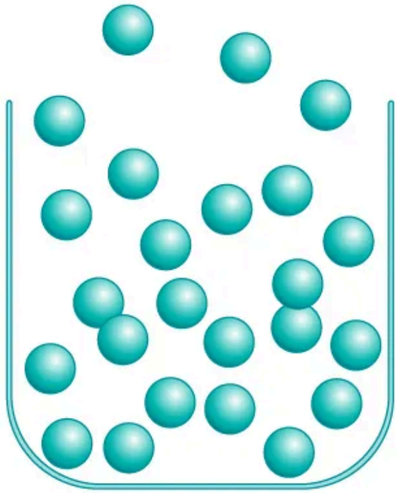
Solid

The molecules that make up a solid are arranged in regular, repeating patterns. They are held firmly in place but can vibrate within a limited area.



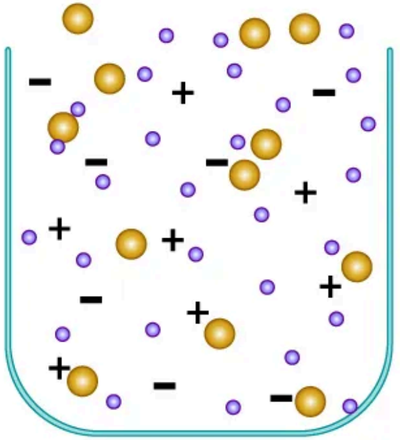
Liquid

The molecules that make up a liquid flow easily around one another. They are kept from flying apart by attractive forces between them. Liquids assume the shape of their containers.



Gas

The molecules that make up a gas fly in all directions at great speeds. They are so far apart that the attractive forces between them are insignificant.



Plasma

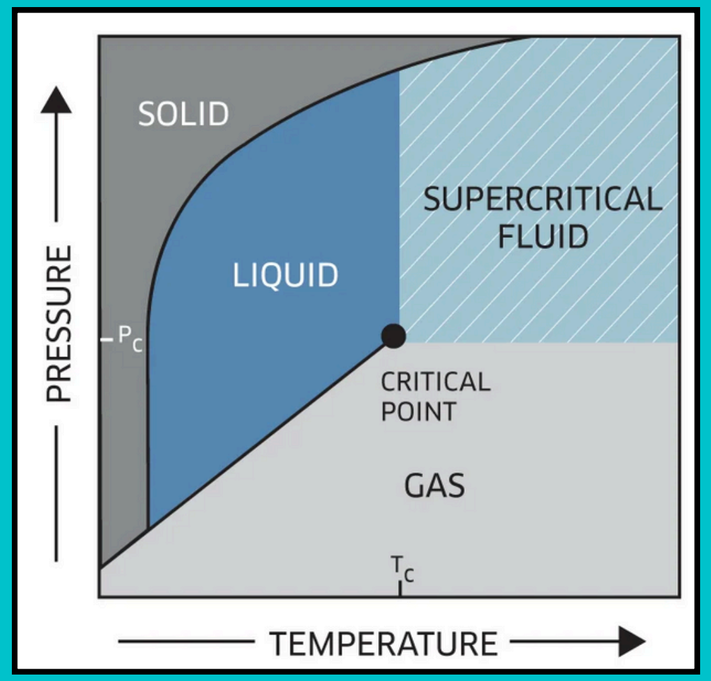
At the very high temperatures of stars, atoms lose their electrons. The mixture of electrons and nuclei that results is the plasma state of matter.

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STATES OF MATTER

Supercritical Fluids

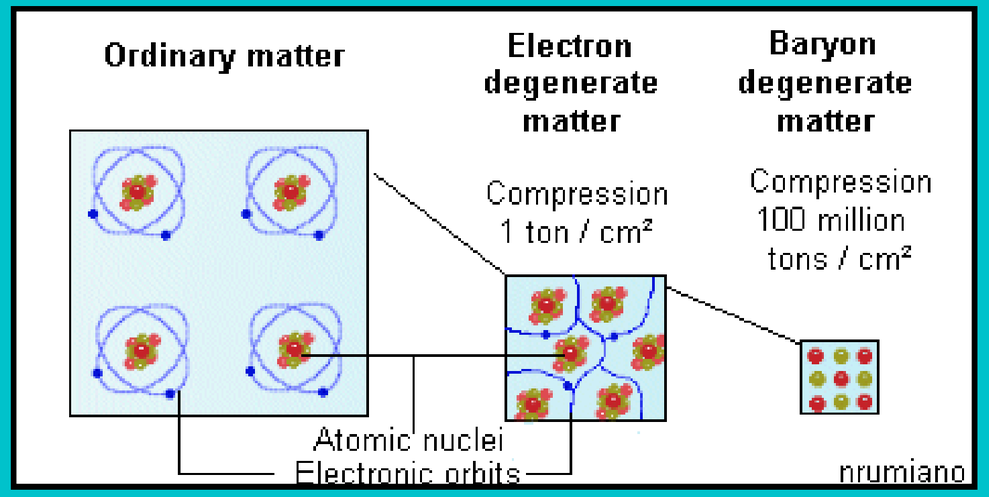
are highly compressed gases which combine properties of gases and liquids.



STATES OF MATTER

Degenerate matter

Is basically a collapsed state of matter. This happens when the atomic structure has broken down because electromagnetic forces are overcome by gravity , this explain the existence of “black holes “white dwarf starts.



GENERAL PROPERTIES OF MATTER





- Matter occupies space.
- Matter has inertia.
- Matter has mass.
- Matter is indestructible.







PROPERTIES OF MATTER



PHYSICAL PROPERTIES

(observable and measurable)

INTENSIVE PROPERTIES

do not depend on the amount of matter.
Examples: color ,odor ,luster ,malleability ,ductility .hardness etc







PROPERTIES OF MATTER

PHYSICAL PROPERTIES

(observable and measurable)



EXTENSIVE PROPERTIES

depend on the amount of matter.
Examples: mass, weight, volume and length



